AGE 310: Instrumentation and Measurements

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Abstract

Lecture 4 transitions to transducers based on electrical energy storage mechanisms: capacitance and inductance. We examine how changes in physical parameters (displacement, material composition) influence these properties. The lecture details the application of capacitive sensors for moisture content (critical in grain storage and soil management) and the high-precision use of the Linear Variable Differential Transformer (LVDT).

Lecture 4: Capacitive and Inductive Transducers

4.1 Capacitive Transducers

4.1.1 Principle of Capacitance

A capacitor is formed by two conductive plates separated by a dielectric (insulating) material. The capacitance (C) is a measure of the ability to store charge. The fundamental relationship is:

$$C = \frac{\varepsilon_0 \varepsilon_r A}{d}$$

Where:

- C: Capacitance (Farads, F)
- ε_0: Permittivity of free space (constant)
- ε_r: Relative Permittivity (Dielectric Constant) of the insulating material
- A: Area of overlap between the plates (m^2)
- d: Distance between the plates (m)

Capacitive transducers work by allowing a physical quantity to vary one of the three geometric or material factors (A, d, ε_r). [Image of Parallel plate capacitor diagram showing Area, Distance, and Dielectric Constant]

4.1.2 Measurement by Varying Distance (d)

If A and ε_r are held constant, capacitance is inversely proportional to distance ($C \propto 1/d$).

- **Applications:** Pressure sensors where a diaphragm deflects under pressure, changing d. This is highly sensitive for small displacements.
- **Limitation:** The output is inherently non-linear, requiring a linearization circuit.

4.1.3 Measurement by Varying Area (A)

If d and ε_r are held constant, capacitance is directly proportional to the overlapping area ($C \propto A$).

- **Applications:** Used for measuring linear displacement (sliding plates) or angular displacement (rotary plates).
- **Ag Application:** Measuring the level of liquid or bulk solids (grain) in a tank or silo. As the level rises, the exposed plate area increases, increasing capacitance.

4.2 Application: Soil and Grain Moisture Sensing

4.2.1 The Variable Dielectric (ε_r)

This is the most crucial application in agricultural engineering. Water has an extremely high dielectric constant ($\varepsilon_r \approx 80$) compared to dry soil ($\varepsilon_r \approx 3-5$) and air ($\varepsilon_r \approx 1$).

When a capacitive probe is inserted into the soil or grain:

$$\text{Total Capacitance} \propto \varepsilon_{\text{soil}} + \varepsilon_{\text{air}} + \varepsilon_{\text{water}}$$

Since water's ε_r dominates, any increase in VWC (Volumetric Water Content) causes a large, measurable increase in the total capacitance.

4.2.2 TDR/FDR Methods

- **FDR (Frequency Domain Reflectometry):** Measures the capacitance of the probe/soil system. The sensor is part of an electronic oscillator circuit. As capacitance (C) increases with moisture, the oscillation frequency (f) decreases. We measure f to determine C and thus VWC.
- **Ag Use:** Smart irrigation scheduling on farms and monitoring post-harvest moisture in grain storage facilities (NCCPB in Morogoro, for example).

Calculation Example 1: Non-Linearity

A capacitive pressure sensor is designed with a plate separation $d = 10$ µm and $A = 1$ cm^2. The material has $\varepsilon_r = 5$. If pressure reduces d to 9 µm (a 10% displacement), calculate the percentage change in capacitance.

Solution: Permittivity Constant (ε_0): 8.85×10^{-12} F/m. Since $C \propto 1/d$, the capacitance is **inversely** related.

$$\frac{C_{\text{new}}}{C_{\text{old}}} = \frac{d_{\text{old}}}{d_{\text{new}}} = \frac{10 \text{ µm}}{9 \text{ µm}} \approx 1.111$$

Percentage change: $(1.111 - 1) \times 100\% = 11.11\%$. A 10% linear change in distance causes an 11.11% change in C, demonstrating the non-linearity.

4.3 Inductive Transducers

4.3.1 Principle of Inductance

Inductive sensors rely on the principle of electromagnetic induction (Faraday's Law). Inductance (L) is the measure of an inductor's opposition to a change in electric current.

$$L = \frac{N^2 \mu_0 \mu_r A}{L_c}$$

Where:

- L: Inductance (Henrys, H)
- N: Number of turns in the coil
- μ_r: Relative Permeability of the core material (Magnetic property)
- A: Cross-sectional area of the core
- L_c: Length of the coil

Inductive transducers measure displacement or proximity by moving a ferromagnetic core (changing μ_r and A) or by changing the mutual inductance between two coils.

4.4 Linear Variable Differential Transformer (LVDT)

4.4.1 Construction and Operation

The LVDT is the gold standard for high-precision, non-contact linear displacement measurement.

- **Primary Coil (P):** Excited by a sinusoidal AC voltage (typically 1 kHz to 10 kHz).
- **Secondary Coils (S_1, S_2):** Two identical secondary coils wound symmetrically, connected in series opposition (anti-phase).
- **Core:** A movable, soft iron core attached to the object being measured.

The output voltage is the difference between the induced voltages in S_1 and S_2: $V_{\text{out}} = V_{S1} - V_{S2}$.

4.4.2 Position Determination

- **Null Position:** Core is perfectly centered. $V_{S1} = V_{S2}$, so $V_{\text{out}} = 0$.
- **Displacement to the Right:** Core couples more flux to S_2. $V_{S2} > V_{S1}$. V_{out} increases and is in phase with the primary voltage.
- **Displacement to the Left:** Core couples more flux to S_1. $V_{S1} > V_{S2}$. V_{out} increases but is 180° out of phase with the primary voltage.

The amplitude of V_{out} gives the magnitude of displacement, and the phase relationship gives the direction.

4.4.3 LVDT Characteristics

- **Pros: Infinite Resolution** (limited only by signal noise), no mechanical friction/wear (non-contact), highly sensitive, and robust.
- **Cons:** Requires complex AC excitation and demodulation circuits; slightly expensive.
- **Ag Use:** Used in precision hydraulic cylinders to provide exact feedback on piston position for tractor auto-steering and hitch depth control.

Calculation Example 2: LVDT Sensitivity

An LVDT has a linearity of $\pm 0.25\%$ over a range of ± 10 mm and a sensitivity of 2.5 V/mm. The bridge is excited at 10 V_{RMS}.

1. Calculate the output voltage when the core is displaced by $+6$ mm.
2. Calculate the total FSO (Full Scale Output) voltage.

Solution:

1. **Output Voltage (V_{out}):**

$$V_{\text{out}} = \text{Sensitivity} \times \text{Displacement} = 2.5 \text{ V/mm} \times 6 \text{ mm} = 15.0 \text{ V}$$

2. **Full Scale Output (FSO):** This is the output at ± 10 mm.

$$\text{FSO} = \text{Sensitivity} \times \text{Max Range} = 2.5 \text{ V/mm} \times 10 \text{ mm} = 25.0 \text{ V}$$

The total swing from negative maximum to positive maximum is 50.0 V.

4.5 Inductive Proximity Sensors

4.5.1 Principle of Operation

These are non-contact switches used for counting, speed detection, or position sensing.

- An internal oscillator creates a high-frequency alternating magnetic field at the sensor tip.
- When a **ferromagnetic material** (metal, e.g., steel bolt) enters the field, eddy currents are induced in the target.
- These eddy currents absorb energy from the oscillator, causing the magnetic field strength to drop dramatically.
- The control circuit detects this drop and switches the output (ON/OFF).

4.5.2 Applications

- **Speed Sensing (RPM):** Used to count the teeth on a rotating gear (e.g., measuring the shaft speed in a thresher or pump).
- **Positioning:** Detecting if a mechanism has reached its limit (e.g., checking if a seeder disc is in the correct position).
- **Ag Use:** Used as non-contact limit switches, and in planter systems to count the rotation of the seed metering unit.

4.6 Summary of Key Terms (Swahili/Context)

- **Capacitance:** Uwezo wa kuhifadhi chaji (Ability to store charge).
- **Dielectric Constant (ε_r):** Kipimo cha kuhifadhi chaji (The measure of material's ability to store charge; very high for water).
- **FDR:** Njia ya FDR (Frequency Domain Reflectometry - Soil moisture sensing method).
- **Inductance:** Uwezo wa kuunda uga sumaku (Ability to induce a magnetic field).
- **LVDT:** Kibadilishaji Tofauti cha Mtiririko (Linear Variable Differential Transformer - High precision displacement sensor).
- **Eddy Currents:** Mikondo ya Eddy (Induced currents in metal used by proximity sensors).
- **Non-Contact:** Bila kugusa (Measurement without physical contact).

Lecture 4: Review Questions

SECTION A: Basic Concepts (5 Questions)

1. State the three physical parameters (A, d, ε_r) that affect the capacitance of a transducer. Provide an example of an agricultural sensor that primarily varies the ε_r parameter.
2. Define the term *Dielectric Constant*. Explain why water makes such a good dielectric for moisture sensing in soil.
3. Briefly describe the construction of an LVDT. How does the output voltage signal indicate both the magnitude and the direction of the core displacement?
4. Explain the principle that causes an Inductive Proximity Sensor to switch ON when a metal object approaches its face.
5. State two distinct advantages of the LVDT over a Potentiometer for measuring displacement in a dusty, high-vibration environment.

SECTION B: Advanced Analysis and Calculations (10 Questions)

6. **Capacitive Pressure Sensor (Advanced Non-Linearity):** A parallel plate capacitive pressure sensor measures a range of 0 to 500 kPa corresponding to a plate separation change from 100 µm to 50 µm.
   - Explain why this sensor is highly non-linear.
   - Calculate the ratio of $C_{\text{max}}/C_{\text{min}}$.
7. **FDR Soil Moisture Analysis:** A new FDR soil moisture probe operating in a circuit outputs a frequency $f_{\text{dry}} = 50$ MHz (dry soil, $\varepsilon_r \approx 4$) and $f_{\text{wet}} = 10$ MHz (saturated soil, $\varepsilon_r \approx 30$). Explain the relationship between ε_r and f. If a soil reading gives $f = 30$ MHz, is the soil closer to dry or saturated conditions?
8. **LVDT Demodulation:** Explain why the LVDT requires AC (Alternating Current) excitation for its primary coil, rather than simple DC. Briefly describe the role of the *Demodulation Circuit* in the LVDT signal conditioning stage.
9. **Inductive Sensor Material Dependence:** Inductive Proximity Sensors are generally designed to sense ferromagnetic metals (like steel) better than non-ferromagnetic metals (like aluminum). Explain the reason for this difference in detection range, based on the principle of the induced magnetic field.
10. **LVDT Calibration:** An LVDT is rated for ± 20 mm displacement and has a sensitivity of 0.5 V/mm. If the output voltage measured is -7.5 V, what is the displacement and direction of the core?
11. **Capacitive Liquid Level Error:** A capacitive sensor measures the level in a fertilizer tank ($\varepsilon_r \approx 5$). Due to heavy rains, a layer of water ($\varepsilon_r \approx 80$) accumulates at the bottom. Explain how this disturbance will affect the sensor's reading and the type of error introduced.

12. **Rotary Capacitive Sensor Design:** An engineer wants to design a rotary capacitive sensor to measure the angular position of a gate valve from $0°$ to $90°$. Describe how they would vary the plate area (A) to achieve a linear relationship between angle and capacitance.

13. **Impedance in Inductive Circuits:** Why is the input impedance of the primary coil of an LVDT important? What would happen if the input frequency of the AC excitation were dramatically reduced from 10 kHz to 1 Hz?

14. **Potentiometer vs. LVDT Trade-off:** Both potentiometers (Lecture 3) and LVDTs measure linear displacement. If the measurement requires high accuracy (± 10 µm) and operates for 10^9 cycles, which sensor is better, and why?

15. **Current Loop Application:** A remote capacitive soil moisture sensor uses an FDR circuit whose frequency is converted into a $4 - 20$ mA output. The output current is 12 mA when the VWC (Volumetric Water Content) is 50%. If the sensor's range is 0% to 100%, calculate the current corresponding to VWC $= 80$%.